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DIGITAL IMAGE PROCESSING

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Abstract

This term paper summarizes key concepts of Digital Image Processing (DIP), demonstrating that the field is based on strong mathematical principles applied via numerical schemes. It covers foundational topics including image representation, acquisition, and digitization. Core techniques in spatial domain enhancement are explored, such as spatial filtering, histogram processing, and image sharpening through derivatives. A significant focus is placed on morphological image processing, detailing the operations of erosion, dilation, opening, and closing. Each section includes concise definitions, essential equations, and notes derived from course lectures.

Keywords: Digital image, Pixel, Bits per pixel (BPP), Sampling, Quantization, Spatial Filtering, Histogram Equalization, Convolution, Smoothing, Sharpening, First-order Derivative, Second-order Derivative, Laplacian, Morphological Operations, Structuring Element, Erosion, Dilation, Opening, Closing, Thresholding.

Contents

1	Introduction	5
2	Image, Digital Image, and BPP	5
2.1	Image and Digital Image	5
2.2	Bits Per Pixel (BPP) and Color Depth	5
3	Image Acquisition and Digitization	6
3.1	Notes on Aliasing	6
4	Representing Digital Images	7
4.1	Color Spaces and Gray Level	7
5	Image Interpolation	7
6	Neighbours and Adjacency	7
7	Enhancement in the Spatial Domain	8
7.1	Spatial Filtering	8
7.1.1	Practical Tip on Boundary Handling	9
8	Image Histogram and Processing Techniques	9
8.1	Histogram Equalization	9
8.2	Local (Adaptive) Histogram Processing	9
9	Spatial Correlation and Convolution	9
10	Smoothing and Sharpening Filters	10
10.1	Smoothing Spatial Filters (Blurring)	10
10.2	Sharpening Spatial Filters	10
10.2.1	First-Order Derivatives	10
10.2.2	Second-Order Derivatives (Laplacian)	11

11 Morphological Image Processing	11
11.1 Structuring Element (SE)	11
11.2 Erosion	12
11.3 Dilation	12
11.4 Opening	12
11.5 Closing	13
11.6 Top-Hat and Bottom-Hat Filters	13
12 Segmentation and Thresholding	13
12.1 Image Binarization and Thresholding	13
13 Conclusion	14

1 Introduction

Digital Image Processing (DIP) refers to the processing of digital images by means of a digital computer. The primary goal is to demonstrate that image processing is founded on robust mathematical principles, which are applied to digital images through various numerical schemes. DIP encompasses a wide range of applications, from the improvement of pictorial information for human interpretation to the processing of image data for autonomous machine perception.

2 Image, Digital Image, and BPP

2.1 Image and Digital Image

An **image** can be defined as a two-dimensional function, $f(x, y)$, where x and y are spatial coordinates, and the amplitude of f at any pair of coordinates (x, y) is called the intensity or gray level of the image at that point.

A **digital image** is a representation of a two-dimensional image as a finite set of digital values. It is composed of a finite number of elements, each with a particular location and value. These elements are referred to by various synonyms depending on the context:

- Pixel (Picture Element) - The most common term.
- Pel
- Sample, spot, or dot.

2.2 Bits Per Pixel (BPP) and Color Depth

The number of distinct colors that can be represented by a pixel depends on the **bits per pixel (bpp)**, also known as color depth. Each additional bit doubles the number of colors available.

- **1 bpp**: $2^1 = 2$ colors (monochrome or binary).

- **8 bpp:** $2^8 = 256$ colors (standard for grayscale).
- **16 bpp:** $2^{16} = 65,536$ colors ("Highcolor"). For this depth, bits are often allocated unevenly across color channels, such as 5 for red, 6 for green, and 5 for blue, because the human eye is more sensitive to variations in green.
- **24 bpp:** $2^{24} \approx 16.7$ million colors ("Truecolor"), with 8 bits for each of the red, green, and blue channels.
- **32 bpp:** Typically uses 24 bits for Truecolor and an additional 8 bits for an **alpha channel**, which describes the pixel's opacity for transparency effects.

3 Image Acquisition and Digitization

Image acquisition is the process of capturing an image, typically using sensors like CCD or CMOS arrays. The output from these sensors is then digitized in a two-step process:

1. **Sampling:** Digitizing the coordinate values. This involves discretizing the continuous spatial coordinates of the image. The sampling rate determines the spatial resolution.
2. **Quantization:** Digitizing the amplitude (intensity) values. This involves mapping the continuous intensity values to a finite set of discrete levels. The number of quantization levels determines the gray-scale resolution.

3.1 Notes on Aliasing

If the sampling frequency is inadequate (i.e., below the Nyquist rate), high-frequency details in the image can be incorrectly represented as lower frequencies, an effect known as aliasing. This can be mitigated by applying an anti-aliasing low-pass filter before sampling.

4 Representing Digital Images

A digital image is commonly represented as a 2D array (matrix) for grayscale images or a 3D array for color images (one 2D array for each color channel).

4.1 Color Spaces and Gray Level

- **RGB:** Represents colors as a combination of Red, Green, and Blue.
- **HSV / HSL:** Represents colors in terms of Hue, Saturation, and Value/Lightness, which is often more intuitive for color-based segmentation.
- **Gray Level:** This term refers to the intensity of monochrome images. A monochromatic image uses only shades of a single color (or gray), ranging from black to white. The only attribute of monochromatic light is its intensity.

5 Image Interpolation

Interpolation is the process of estimating image values at non-grid positions, which is essential for tasks like resizing, rotation, and warping.

- **Nearest-neighbor:** Assigns the value of the nearest known pixel. It is fast but can produce blocky artifacts.
- **Bilinear:** Performs linear interpolation in two dimensions using the four nearest neighbors. It produces smoother results.
- **Bicubic:** Uses a 4×4 neighborhood and cubic polynomials to estimate the value. It is computationally more expensive but preserves finer detail and produces even smoother results.

6 Neighbours and Adjacency

- **4-neighbours ($N_4(p)$):** For a pixel p at (x, y) , its 4-neighbours are at $(x \pm 1, y)$ and $(x, y \pm 1)$.

- **8-neighbours** ($N_8(p)$): Includes the 4-neighbours plus the four diagonal neighbours at $(x \pm 1, y \pm 1)$.
- **Adjacency**: Two pixels are considered adjacent if they are neighbours and their gray levels satisfy a specified criterion (e.g., they belong to the same set of intensity values, V).

7 Enhancement in the Spatial Domain

Enhancement is the process of manipulating an image to make it more suitable for a specific application. The spatial domain refers to the image plane itself, where operations are performed directly on the pixel values. The general expression for spatial domain processing is:

$$g(x, y) = T[f(x, y)]$$

where $f(x, y)$ is the input image, $g(x, y)$ is the output image, and T is an operator defined over a neighborhood of (x, y) .

7.1 Spatial Filtering

Spatial filtering involves moving a filter mask (also called a kernel, template, or window) over the image. At each position, a new value for the center pixel is calculated based on the values of the pixels in its neighborhood and the coefficients of the filter mask. itemize

Linear Filters: If the operation is linear (e.g., a sum of products, like convolution), the filter is a linear spatial filter.

Non-linear Filters: If the operation is non-linear (e.g., taking the median or maximum), it is a non-linear spatial filter.

7.1.1 Practical Tip on Boundary Handling

When the filter mask is near the image border, some of its elements may fall outside the image. Common strategies to handle this include zero-padding, replicating the border pixels, or reflecting the image at the boundary.

8 Image Histogram and Processing Techniques

An image histogram is a plot that shows the frequency of each intensity level in an image. A normalized histogram, where each frequency is divided by the total number of pixels, can be interpreted as a probability distribution $p(r_k)$ for each intensity level r_k .

8.1 Histogram Equalization

This technique aims to produce an output image with a uniform histogram, thereby increasing the global contrast of the image. The transformation function is based on the cumulative distribution function (CDF) of the input image's histogram. The transformation $s = T(r)$ must be single-valued and monotonically increasing to prevent intensity reversals.

8.2 Local (Adaptive) Histogram Processing

While global equalization uses the entire image, local processing enhances details by applying histogram-based transformations to smaller neighborhoods. A neighborhood is defined and moved across the image, and at each location, a local histogram is computed and used to transform the value of the center pixel. This can enhance local contrast but may also amplify noise.

9 Spatial Correlation and Convolution

Correlation and convolution are fundamental operations in linear spatial filtering. For a 2D image f and a kernel h :

- **Correlation:** $(f \star h)[i, j] = \sum_m \sum_n f[m, n]h[i + m, j + n]$
- **Convolution:** $(f * h)[i, j] = \sum_m \sum_n f[m, n]h[i - m, j - n]$

Convolution is equivalent to correlation if the filter kernel is flipped horizontally and vertically. For symmetric kernels, they are identical.

10 Smoothing and Sharpening Filters

10.1 Smoothing Spatial Filters (Blurring)

Smoothing is used to reduce noise and blur fine detail. It is analogous to **integration**, as it involves averaging pixel values in a neighborhood.

- **Box (Mean) Filter:** A simple filter where all coefficients are equal. It replaces each pixel with the average of its neighbors.
- **Gaussian Filter:** Uses a kernel where weights decrease with distance from the center, resulting in a smoother blur.
- **Median Filter:** A non-linear filter that replaces a pixel with the median value in its neighborhood. It is highly effective against salt-and-pepper (impulse) noise.

10.2 Sharpening Spatial Filters

Sharpening aims to highlight fine detail and enhance blurred edges. It is analogous to **spatial differentiation**. The strength of a derivative operator's response is proportional to the degree of intensity discontinuity at that point. Thus, differentiation enhances edges and noise while de-emphasizing areas with slowly varying intensities.

10.2.1 First-Order Derivatives

The gradient $\nabla f = [\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}]$ points in the direction of the greatest rate of change. Its magnitude $|\nabla f|$ is used to detect edges. Discrete approximations include the Sobel, Prewitt, and Roberts operators.

10.2.2 Second-Order Derivatives (Laplacian)

The second derivative is used to measure curvature and is excellent for locating edges, often identified by its zero-crossings. The Laplacian is an isotropic second-derivative operator:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Using finite differences, this can be expressed as:

$$\nabla^2 f = [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1)] - 4f(x, y)$$

This corresponds to a filter kernel such as:

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

An **enhanced Laplacian** filter, used for sharpening, is created by subtracting the Laplacian from the original image. This can be implemented with a single kernel by adding 1 to the center coefficient (e.g., changing -4 to -5).

11 Morphological Image Processing

Morphology is a collection of non-linear operations related to the shape or structure of features in an image. It is particularly suited for binary images and is used to remove imperfections like noise and texture after processes like thresholding.

11.1 Structuring Element (SE)

Morphological techniques probe an image with a small template called a **structuring element (SE)**. The SE is a small binary matrix of 0s and 1s that defines a specific shape (e.g., square, disk, line) and size. The SE is positioned at all possible locations in the image and compared with the corresponding neighborhood of pixels.

11.2 Erosion

Erosion ($A \ominus B$) shrinks objects by stripping away a layer of pixels from their boundaries. For a binary image, the output pixel is set to 1 only if the SE, when centered at that pixel, **fits** entirely within the foreground of the input image.

- **Effect:** It decreases object sizes, removes small details, and enlarges holes.
- **Grayscale:** It computes the local minimum over the neighborhood defined by the SE.

11.3 Dilation

Dilation ($A \oplus B$) grows objects by adding a layer of pixels to their boundaries. The output pixel is set to 1 if the SE, when centered at that pixel, **hits** (intersects with) the foreground of the input image.

- **Effect:** It increases object sizes, fills small holes, and connects nearby objects.
- **Grayscale:** It computes the local maximum over the neighborhood.
- **Duality:** Dilation and erosion are dual operations. The erosion of a foreground is equivalent to the dilation of its background.

11.4 Opening

Opening ($A \circ B$) is defined as an erosion followed by a dilation with the same SE:

$$A \circ B = (A \ominus B) \oplus B$$

Opening smooths object contours, breaks thin connections, and removes small protrusions. It is called "opening" because it can open up a gap between objects connected by a thin bridge. It is an idempotent operation.

11.5 Closing

Closing ($A \bullet B$) is defined as a dilation followed by an erosion with the same SE:

$$A \bullet B = (A \oplus B) \ominus B$$

Closing tends to fuse narrow breaks, fill small holes, and connect nearby objects.

11.6 Top-Hat and Bottom-Hat Filters

- **Top-Hat Filter:** Subtracts the opened image from the original image: $T_{hat}(A) = A - (A \circ B)$. It is used to extract small, bright features against a darker background.
- **Bottom-Hat Filter:** Subtracts the original image from the closed image: $B_{hat}(A) = (A \bullet B) - A$. It is used to extract small, dark features.

12 Segmentation and Thresholding

Segmentation is the process of partitioning an image into a collection of distinct regions, where each region contains pixels with similar attributes (e.g., intensity, color, texture). Meaningful segmentation is a critical first step from low-level image processing to higher-level image analysis.

12.1 Image Binarization and Thresholding

Binarization is the process of converting a grayscale image into a binary image. The most common method is **global thresholding**, where a single threshold value, T , is chosen. Each pixel is then classified as foreground or background:

$$g(i, j) = \begin{cases} 1 & \text{if } I(i, j) \geq T \\ 0 & \text{if } I(i, j) < T \end{cases}$$

An optimal threshold often aims to create large connected regions while minimizing small, artifactual regions. **Otsu's method** is a popular automatic technique that chooses a threshold by minimizing the intra-class intensity variance.

13 Conclusion

This paper has provided a summary of fundamental Digital Image Processing topics, incorporating core principles from lectures. We began with the basic definitions of digital images and their representation and moved to spatial domain techniques. Key concepts of enhancement, including histogram-based methods and spatial filtering for smoothing and sharpening, were detailed. The role of first and second-order derivatives, particularly the Laplacian, in edge detection and sharpening was explained. A significant portion was dedicated to morphological processing, a powerful non-linear approach for analyzing and modifying image structures using erosion, dilation, and their combinations. These foundational techniques form the building blocks for more advanced image analysis and machine perception tasks.